

EXPERIMENT 9

Student Name: Shagun Goyal
Branch: AIT-CSE
Subject Name: Computer Network

UID: 24BDA70214
Section/Group: 24AIT_KRG-G2
Subject Code: 24CSH-337

Experiment 9– Implement Connection using TCP

Objective

- To study the fundamental working principles and important features of UDP (User Datagram Protocol).
- To implement communication between a UDP client and server using Python socket programming.
- To establish message exchange between server.py and client.py through UDP sockets.
- To understand how data can be transmitted and received as datagrams without creating a dedicated connection between the client and server.

Theory

UDP (User Datagram Protocol) is a connectionless transport-layer protocol that enables devices to exchange data over a network. Unlike TCP, UDP does not establish a connection before sending data and does not provide guarantees for packet delivery, sequence, or retransmission.

In UDP communication, information is transmitted in the form of independent **datagrams**. Each datagram is sent to a specified destination using its IP address and port number.

In this experiment, Python's built-in **socket module** is used to create a UDP-based client-server application. The server creates a UDP socket and associates it with a particular IP address and port using the `bind()` function. The client creates its socket and transmits messages to the server using UDP communication. The server receives incoming datagrams using `recvfrom()` and can send a response back to the client using `sendto()`.

Since UDP is a connectionless protocol, it does not require a three-way handshake or any connection establishment procedure. As a result, it generally provides faster communication with lower overhead. UDP is commonly used in applications where speed and low latency are important, such as **online gaming, video/audio streaming, DNS, and real-time communication**.

Practical Steps

1. Open a project directory in a code editor such as **VS Code, PyCharm, or IntelliJ IDEA**.
2. Create two Python files named `server.py` and `client.py`.
3. Enter the server-side program in `server.py` and the client-side program in `client.py`.
4. Save both Python files.

EXPERIMENT 9

5. Open two separate terminals or use a split-terminal view.
6. In **Terminal 1**, execute the server program using:
python server.py
7. In **Terminal 2**, execute the client program using:
python client.py
8. Enter a message from the client side through the Terminal 2 command line.
9. Enter a response from the server side through Terminal 1.
10. To terminate the communication, enter **"Over"** from the client side.

Code

Server.py

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
server_socket.bind(("127.0.0.1", 5000))
print("UDP Server started")
print("Waiting for messages...")
while True:
    message, client_address = server_socket.recvfrom(1024)
    message = message.decode()
    print("Client:", message)
    if message == "Over":
        break
    reply = input("Enter reply: ")
    server_socket.sendto(reply.encode(), client_address)
server_socket.close()
print("Server closed")
```

EXPERIMENT 9

Client.py

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

client_socket.connect(("127.0.0.1", 5000))

print("UDP Client started")

while True:

    message = input("Enter message: ")

    client_socket.send(message.encode())

    # encode(): Converts a string into bytes

    if message == "Over":

        break

    reply = client_socket.recv(1024).decode()

    print("Server:", reply)

client_socket.close()

print("Client closed")
```

Result

<pre>PS D:\University\Year 3\CN\UDP_Socket_Experiment\UDP_Socket_Experiment> python client.py UDP Client started Enter message: Hi!!, Shagun Server: You deserve 30/30 marks. Good Child Enter message: Thank you so much. Server: Mention not beta, yeh low 30 / 30 in all experiments Enter message: Over Client closed</pre>	<pre>PS D:\University\Year 3\CN\UDP_Socket_Experiment\UDP_Socket_Experiment> python server.py Waiting for a client... Client: Hi!!, Shagun Enter reply: You deserve 30/30 marks. Good Child Client: Thank you so much. Enter reply: Mention not beta, yeh low 30 / 30 in all experiments Client: Over Connection closed</pre>
--	--

Learning Outcome

- Gained an understanding of the working mechanism and key characteristics of UDP and its distinction from connection-oriented protocols such as TCP.
- Learned how to create and execute a UDP client-server application using Python socket programming.
- Understood how datagrams are transmitted and received without establishing a dedicated connection.